6. Identify the errors and correct them.

(i) $7y^2 - y^2 + 3y^2 = 10y^2$ (ii) $6xy + 3xy = 9x^2y^2$

(iii) $m(4m-3) = 4m^2 - 3$ (iv) $(4n)^2 - 2n + 3 = 4n^2 - 2n + 3$

(v) $(x-2)(x+3) = x^2 - 6$ (vi) $-3p^2 + 4p - 7 = -(3p^2 + 4p - 7)$

7. Statement A: If $24p^2q$ is divided by $3pq$, then the quotient is $8p$.

Statement B: Simplification of $\frac{(5x+5)}{5}$ is $5x$.

(i) Both A and B are true (ii) A is true but B is false

(iii) A is false but B is true (iv) Both A and B are false

8. Statement A: $4x^2 + 3x - 2 = 2(2x^2 + \frac{3x}{2} - 1)$

Statement B: $(2m-5)-(5-2m) = (2m-5) + (2m-5)$

(i) Both A and B are true (ii) A is true but B is false

(iii) A is false but B is true (iv) Both A and B are false

3.5 Identities

We have studied in the previous class about standard algebraic identities. An identity is an equation satisfied by any value that replaces its variable(s). Now, we shall recollect four known identities, which are,

$$(a+b)^2 \equiv a^2 + 2ab + b^2 \qquad (a-b)^2 \equiv a^2 - 2ab + b^2$$

$$(a^2 - b^2) \equiv (a+b)(a-b) \qquad (x+a)(x+b) \equiv x^2 + (a+b)x + ab$$

Instead of the symbol ≡, we use = to represent an identity without any confusion.



Try these

Expand the following

(i) $(p+2)^2 =$ ..... (ii) $(3-a)^2 =$ .....

(iii) $(6^2 - x^2) =$ ..... (iv) $(a+b)^2 - (a-b)^2 =$ .....

(v) $(a+b)^2 = (a+b) \times$ ..... (vi) $(m+n)(.....) = m^2 - n^2$

(vii) $(m+...)^2 = m^2 + 14m + 49$ (xiii) $(k^2 - 49) = (k+\ldots)(k-\ldots)$

(ix) $m^2 - 6m + 9 =$ ..... (x) $(m-10)(m+5) =$ .....



Note

x=1 is the only solution for $7x + 3 = 10$ whereas any value of x satisfies $(x+2)^2 = x^2 + 4x + 4$. So $7x + 3 = 10$ is an equation $(x+2)^2 = x^2 + 4x + 4$ is an identity. An identity is an equation but vice versa is not true.

3.5.1 Application of Identities

The identities give an alternative method of solving problems on multiplication of algebraic expressions and also of numbers.

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